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# A Safety Detection Method for Electric Bike Riding Incorporating License Plate Region Recognition

SHAOHUI ZHONG<sup>1</sup>, and XIAOFEI LIU<sup>2,3</sup>

<sup>1</sup>Software College of Changsha Institute of Technology, Changsha Institute of Technology, Changsha, Hunan, China

<sup>2</sup>School of Artificial Intelligence, Guangdong Open University, Guangzhou, Guangdong, China

<sup>3</sup>Faculty of Data Science, City University of Macau, Macau, China

Corresponding author: Xiaofei Liu (e-mail: lxf@gdrtvu.edu.cn).

**ABSTRACT** The widespread use of electric bikes has made rider safety a crucial issue in intelligent transportation research. To address the limitations of existing methods in terms of accuracy and robustness in electric bike license plate recognition, helmet detection, and rider detection tasks, this paper proposes a novel electric bike safety detection method, F-YOLOv8, which integrates license plate region recognition. Based on the YOLOv8 architecture, this method incorporates a feature pyramid network (FPN) to optimize multi-scale object detection capabilities, while introducing an attention mechanism to enhance the extraction of key features. The study constructs a high-quality dataset covering various riding scenarios and violations, including image data collected in Guangzhou and Foshan, Guangdong Province, annotated using LabelImg. The label types include Ebike (rider), plate (license plate), gzplace (license plate region), helmet (wearing a helmet), nohelmet (not wearing a helmet), and manned (carrying a passenger). Experimental results demonstrate significant improvements across all metrics: the precision on Ebike reached 96.1%, on Plate reached 92.9%, on Helmet reached 84.6%, and on Manned tasks reached 67.9%. This study provides an efficient and reliable detection method for enhancing electric bike riding safety management, laying a technical foundation for the development of intelligent transportation systems.

**INDEX TERMS** Electric bike safety detection, License plate recognition, Helmet detection, Rider detection, YOLO

## I. INTRODUCTION

With the acceleration of urbanization and the improvement of people's living standards, electric bikes (hereinafter referred to as e-bikes) have become an important mode of transportation. Their convenience and affordability have led to a rapid increase in the number of e-bikes, especially in developing countries such as China [1]. However, the safety issues related to e-bike riding have become increasingly prominent, with frequent traffic accidents causing significant loss of life, property, and public safety concerns. According to the "China Statistical Yearbook 2024," the annual growth rate of fatalities in traffic accidents involving e-bikes from 2019 to 2023 reached 5.85%, highlighting the seriousness of e-bike accidents.

The causes of e-bike accidents are complex and diverse, with one of the main factors being traffic violations. For example, behaviors such as not wearing helmets, carrying passengers illegally, and running red lights significantly in-

crease the risk of accidents [2]. Additionally, due to the small size and high speed of e-bikes, collisions are more likely to occur in complex traffic environments. Therefore, how to effectively supervise and regulate e-bike riding behavior, and improve riding safety, has become a pressing social issue.

In recent years, the rapid development of computer vision technology and deep learning has provided new solutions for intelligent traffic management [3]. Object detection technology, as one of the core tasks in the field of computer vision, can automatically recognize and locate objects in images or videos, such as vehicles, pedestrians, etc. [4]. Applying object detection technology to e-bike riding safety monitoring allows for the automatic identification of violations, such as helmet-wearing detection and passenger-carrying detection. Additionally, license plate recognition technology can help identify violating vehicles and trace their travel trajectories, providing strong support for traffic management [5].

While progress has been made in the research of e-bike

riding safety detection methods, many challenges remain. Traditional image processing-based technologies, such as those discussed by Park et al. [6], indicate that Haar features, when used for object detection, suffer from limited feature representation ability due to manual design, resulting in poor robustness and accuracy in complex scenarios (e.g., lighting changes, occlusion). Although the application of deep learning, as introduced by Wang et al. [7], significantly improves detection performance in license plate recognition compared to traditional methods, precision and recall still face bottlenecks, particularly for small license plates that are prone to background interference in complex environments. Furthermore, the limitations in computational resources make it difficult for deep learning models to perform real-time detection on resource-constrained edge devices. More importantly, while some studies have explored tasks such as license plate recognition [8]–[10] and helmet-wearing detection [11] [12], joint detection methods for e-bike riding safety are still scarce. Treating these tasks independently separates their potential interconnections. For instance, combining helmet-wearing status, passenger-carrying information, and license plate data could provide more comprehensive information for identifying violations and assessing riding safety. Therefore, developing a multi-task joint detection method for e-bike riding safety that integrates multiple sources of information while balancing detection accuracy and real-time performance is crucial.

To address the limitations of existing electric bike riding safety detection methods, this paper proposes a multi-task joint detection framework [13] that incorporates license plate region recognition, named F-YOLOv8. The main contributions of this work are as follows:

- We design an improved YOLOv8 architecture enhanced with Transformer and Feature Pyramid Network (FPN) modules to improve multi-object detection accuracy, particularly for small targets such as license plates.
- We construct a large-scale dataset covering diverse riding scenarios and violation types, providing a valuable resource for future research in electric bike safety.
- We experimentally demonstrate the superior performance of F-YOLOv8 in terms of mAP, precision, and inference speed, validating its effectiveness and potential for real-world deployment.

## II. RELATED WORK

### A. OBJECT DETECTION ALGORITHMS

Object detection is one of the fundamental tasks in the field of computer vision, aiming to identify the location and category of specific objects in images or videos. Traditional object detection methods are typically based on manually designed features and sliding window mechanisms, such as the Viola-Jones detector [16] and HOG+SVM [17]. However, these methods have poor adaptability to complex scenarios and are computationally expensive. In recent years, with the rise of deep learning, significant advancements have been made in deep learning-based object detection algorithms. These

methods can automatically learn image features and achieve end-to-end object detection. Among them, two-stage detectors based on convolutional neural networks (CNNs), such as Faster R-CNN [18], achieve higher detection accuracy by generating candidate regions and performing classification and regression. However, two-stage detectors have higher computational complexity, making it challenging to meet real-time requirements. To improve detection speed, single-stage detectors, such as SSD [19] and the YOLO series [20], directly predict the location and category of objects, achieving a good balance between speed and accuracy. Due to their fast and efficient nature, YOLO-based algorithms are widely used in practical applications and have become a key choice for real-time object detection.

### B. AUTOMATIC LICENSE PLATE RECOGNITION

Automatic License Plate Recognition (ALPR) technology refers to the automated identification of vehicle license plate numbers and their associated regions. Traditional license plate recognition methods typically include steps such as image preprocessing, license plate localization, character segmentation, and character recognition [21]. These methods often rely on manually designed features and rules, making them sensitive to factors like lighting, angles, and noise. In recent years, deep learning-based license plate recognition methods have gradually become mainstream [22]. These methods treat license plate recognition as an object detection and recognition task, enabling direct detection and recognition of license plate information from images. For instance, Singh et al. [23] and Dalarmelina et al. [24] both studied using deep learning models for license plate localization, followed by optical character recognition (OCR) technology for character recognition. Research by Al-Batat et al. [25] and Qin et al. [26] treats license plate recognition as an end-to-end task, directly using deep learning models to predict the license plate number. These deep learning-based methods outperform traditional methods in terms of robustness and accuracy.

### C. DETECTION OF HELMET WEARING AND PASSENGER PRESENCE

Helmet-wearing detection and passenger detection are essential components of e-bike riding safety monitoring. Existing helmet-wearing detection methods primarily rely on image processing technologies and machine learning algorithms. Some methods detect helmets based on features such as color, shape, and texture. For example, Wu et al. [27] proposed a helmet recognition method based on intelligent vision, which focuses on monitoring whether workers are wearing helmets and identifies the color of the helmets. This method was tested on a dataset, and the average accuracy of helmet recognition was 90.3%. Other methods use machine learning algorithms to classify helmets, such as the innovative and practical helmet-wearing detection method based on image processing and machine learning proposed by Li et al. [28], which demonstrated the effectiveness of the approach.

Passenger detection can be considered a specific type of pedestrian detection or human detection problem, so general object detection techniques such as Faster R-CNN [18] and YOLO [20] can be used to detect whether there is a human in the rear seat area of the e-bike, thereby determining if there is a passenger. However, due to the relatively small size of the e-bike's rear seat area and the varied postures of passengers, accurately detecting the presence of a passenger remains a significant challenge.

#### D. MTL

MTL is a machine learning paradigm that aims to improve the generalization ability of models by simultaneously learning multiple related tasks [29]. In the field of object detection, MTL can combine object detection with other related tasks, such as object classification and semantic segmentation, to train the model, thereby enhancing the overall performance [30]. Some studies have shown that MTL can improve the accuracy and robustness of object detection [31]–[33].

In summary, existing methods in object detection, license plate recognition, helmet-wearing detection, and passenger detection have made progress in their respective fields. However, integrating these tasks into a unified framework to achieve efficient and accurate e-bike riding safety detection remains a challenge. This study aims to address this challenge by proposing an e-bike riding safety detection method that integrates license plate region recognition to improve detection accuracy and efficiency.

### III. YOLOV8 NETWORK STRUCTURE

YOLO is an end-to-end deep learning-based object detection algorithm that formulates the detection task as a regression problem, enabling direct output of both bounding boxes and class information. Compared with conventional two-stage detection approaches, YOLO exhibits faster detection speed, meeting the real-time requirements for electric bike riding. Its model structure is simple yet efficient, offering strong scalability. Furthermore, it achieves relatively high accuracy in detecting small objects, which is particularly advantageous for recognizing such targets in images captured under complex conditions. By employing a single end-to-end CNN architecture, YOLO performs predictions directly on the input image, eliminating the need for a separate candidate region generation phase followed by classification, as in R-CNN. This design significantly accelerates detection, simplifies the training process, and can be readily optimized and enhanced for remote sensing applications.

Unlike previous versions of YOLO, YOLOv8 introduces a modular network architecture that adapts to different scales and application requirements. Significant improvements and upgrades have been made in both the algorithmic structure and training strategies, with the adoption of an adaptive training approach that dynamically adjusts training parameters based on varying input resolutions and tasks. To accommodate a wide range of application scenarios, YOLOv8 offers multiple versions, including YOLOv8n,

TABLE 1. YOLOv8 network model parameters

| Network Model | Network Depth | Network Width |
|---------------|---------------|---------------|
| YOLOv8n       | 0.33          | 0.25          |
| YOLOv8s       | 0.33          | 0.50          |
| YOLOv8m       | 0.67          | 0.75          |
| YOLOv8l       | 1.00          | 1.00          |
| YOLOv8x       | 1.00          | 1.00          |

YOLOv8s, YOLOv8m, YOLOv8l, and YOLOv8x. The differences among these versions primarily lie in variations in the model structure, specifically the adjustment of depth and width factors. These adjustments are achieved by modifying the number of layers and channels in each layer. Furthermore, by setting different depth levels and kernel quantities in each Cross-Stage Partial Network (C2f) module, YOLOv8 generates networks with varying depths and widths.

This paper uses YOLOv8n as the base network for electric bike riding safety detection. The YOLOv8 network architecture consists of four main components: Input, Backbone, Neck, and Head, as shown in Fig.1. Specifically, Conv refers to the basic convolutional unit, C2f is a cross-stage local network residual block containing multiple convolution modules, Concat represents the feature map fusion operation, Upsample refers to the upsampling operation, and Bottleneck is the bottleneck layer designed to optimize computational efficiency.

### IV. TRANSFORMER ARCHITECTURE

Transformer is a neural network architecture originally proposed for sequence modeling, characterized by its self-attention mechanism and parallel processing capabilities. Unlike traditional RNN-based models, Transformer enables global context modeling through multi-head attention, making it suitable for capturing complex feature dependencies in visual tasks.

In our method, we adopt the self-attention structure from the Transformer encoder to enhance feature representation. Specifically, each attention head computes a weighted sum of value vectors (V), where the weights are determined by the similarity between the query (Q) and key (K) vectors. The multi-head mechanism allows the model to attend to information from different representation subspaces.

A high-level overview of the Transformer encoder is shown in Fig. 2, which consists of stacked self-attention and feed-forward layers with residual connections and layer normalization.

### V. DETR ARCHITECTURE

DETR is a novel deep learning model for object detection tasks that combines the strengths of Transformer and convolutional networks. Unlike traditional models, which rely on region proposals or anchor boxes, DETR treats object detection as a set prediction problem. This allows the model

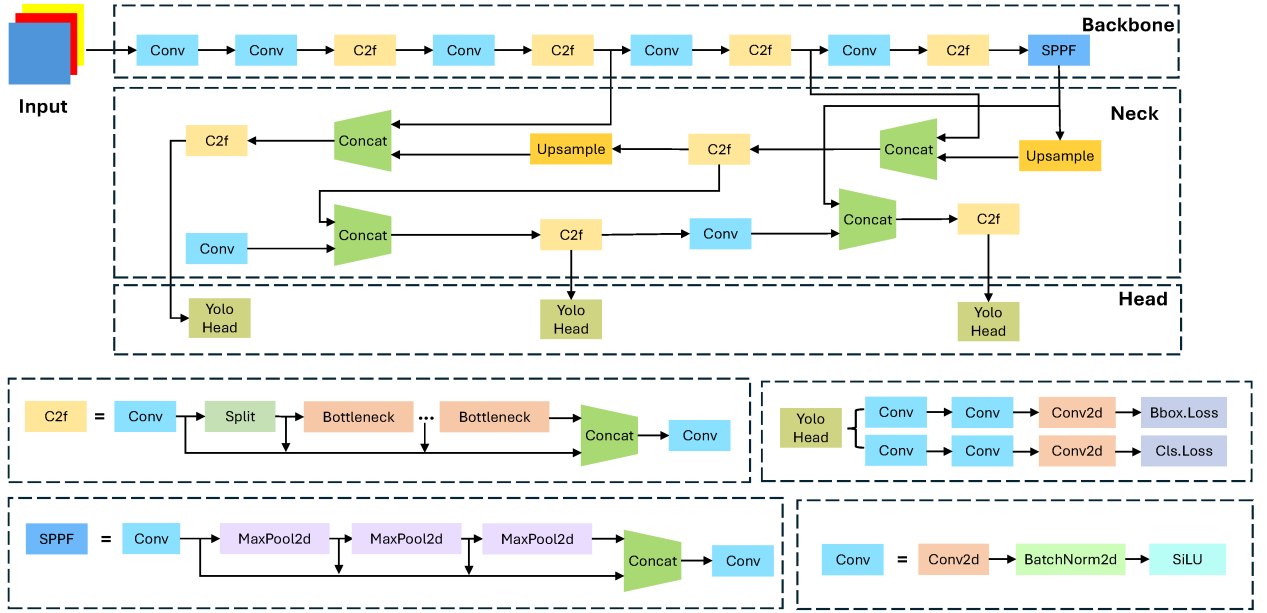


FIGURE 1. Structure of the YOLOv8 model.

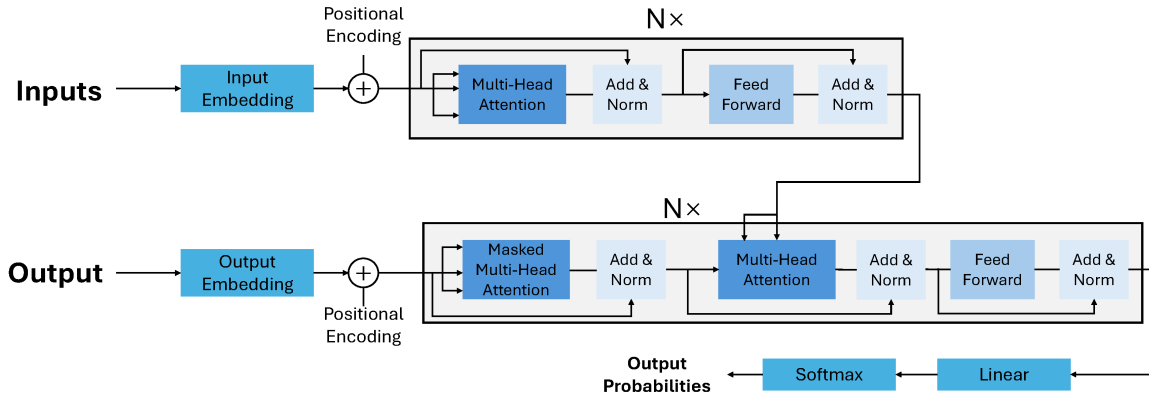


FIGURE 2. Structure of transformer.

to predict the locations and classes of objects directly from the image's feature map, making it end-to-end trainable.

DETR frames object detection as a sequence-to-sequence task and uses a set-based global loss function to match predicted object positions with ground truth boxes. This enables the model to predict all objects in parallel, overcoming the limitations of traditional models in handling complex scenes. Leveraging the sequential structure of the transformer model, DETR converts image sequences into sets, which are then visualized using learnable positional embeddings. The architecture of the DETR network is shown in Fig.3.

#### A. INPUT PROCESSING AND BACKBONE NETWORK

DETR starts by extracting features from the input image using a convolutional backbone, such as ResNet. The backbone

produces a feature map of size  $H \times W \times C$ , where  $H$  and  $W$  are the spatial dimensions, and  $C$  is the number of channels. This feature map is fed into the Transformer for further processing.

#### B. POSITIONAL ENCODING

Since Transformer does not inherently capture spatial relationships between image regions, DETR introduces positional encoding to provide information about each region's position in the image. The positional encoding is added to the feature map to help the Transformer understand spatial dependencies.

#### C. TRANSFORMER ENCODER AND DECODER

The encoder processes the feature map, capturing both local and global dependencies through self-attention and feed-



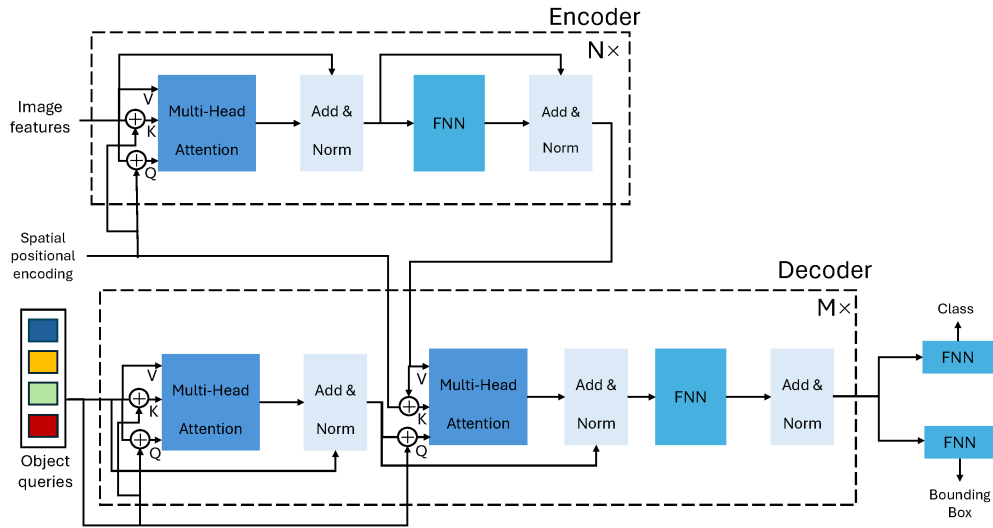


FIGURE 3. Structure of DETR.

forward networks. The decoder uses fixed query embeddings to predict object locations and classes, attending to the encoder's output for contextual information.

#### D. PREDICTION AND OUTPUT

The decoder's output is passed through a classification layer and a regression layer. The classification layer predicts the object class, while the regression layer predicts the bounding box coordinates. These predictions are compared to the ground truth using a bipartite matching loss:

$$\text{BBox}_i = \text{Regressor}(Q_i). \quad (1)$$

DETR simplifies the object detection pipeline by predicting all objects in parallel and directly matching predicted boxes with ground truth, making it an efficient and robust solution for complex object detection tasks.

Since the sequence of output objects does not align with the ground-truth sequence, DETR employs a binary matching technique to map the Ground Truth (GT) frames to the predicted frames. The matching strategy is outlined as follows:

$$\hat{\sigma} = \underset{\sigma \in \mathbb{S}_N}{\operatorname{argmin}} \sum_i^N \mathcal{L}_{\text{match}}(y_i, \hat{y}_{\sigma(i)}). \quad (2)$$

$$\mathcal{L}_{\text{match}}(y_i, \hat{y}_{\sigma(i)}) = -1_{\{c_i \neq \emptyset\}} \hat{p}_{\sigma(i)}(c_i) + 1_{\{c_i \neq \emptyset\}} \mathcal{L}_{\text{box}}(b_i, \hat{b}_{\sigma(i)}). \quad (3)$$

where  $y_i$  represents the true target,  $\hat{y}_{\sigma(i)}$  denotes the predicted target,  $b_i$  is the true bounding box, and  $\hat{b}_{\sigma(i)}$  is the predicted bounding box.

The final loss function is:

$$\mathcal{L}_{\text{Hungarian}}(y, \hat{y}) = \sum_{i=1}^N \left[ -\log \hat{p}_{\hat{\sigma}(i)}(c_i) + 1_{\{c_i \neq \emptyset\}} \mathcal{L}_{\text{box}}(b_i, \hat{b}_{\hat{\sigma}(i)}) \right]. \quad (4)$$

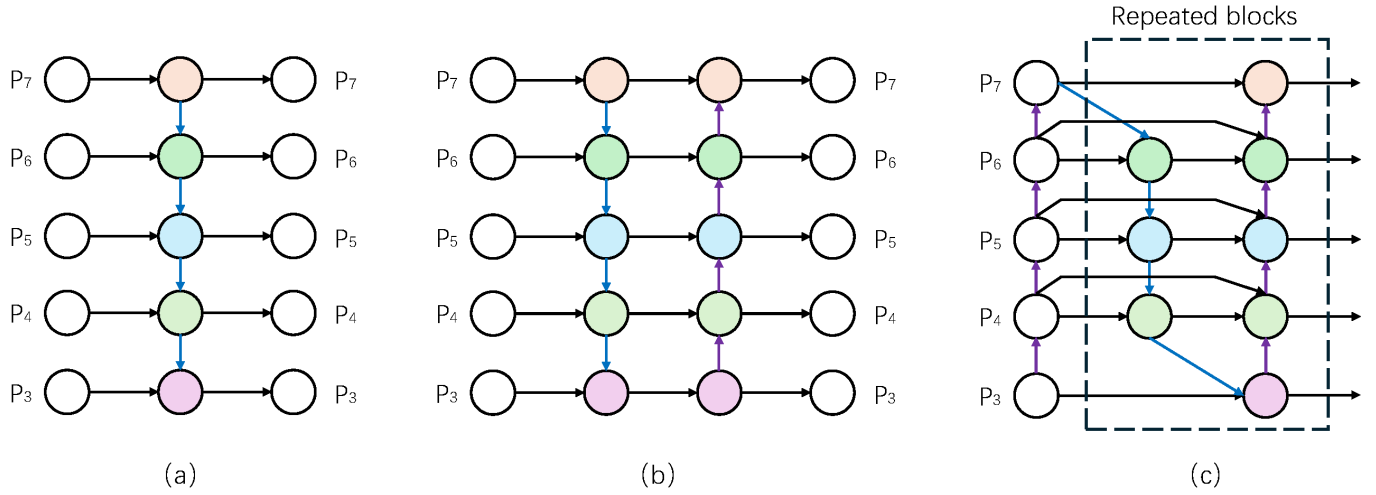
$$\mathcal{L}_{\text{box}}(b_i, \hat{b}_{\sigma(i)}) = \lambda_{\text{iou}} \mathcal{L}_{\text{iou}}(b_i, \hat{b}_{\sigma(i)}) + \lambda_{L1} \|b_i - \hat{b}_{\sigma(i)}\|_1. \quad (5)$$

Here,  $\mathcal{L}_{\text{iou}}$  is the Generalized Intersection over Union (GIoU) loss,  $\lambda_{\text{iou}}$  is the weight factor for the IoU loss,  $\lambda_{L1}$  is the weight factor for the L1 loss,  $b_i$  is the true bounding box,  $\hat{b}_{\sigma(i)}$  is the predicted bounding box, and  $\hat{p}_{\sigma(i)}(c_i)$  is the probability that the target belongs to a particular category. Therefore, the DETR algorithm can effectively recognize the images involved in this paper.

#### VI. IMPROVED YOLOV8 DETECTION ALGORITHM

In this paper, we propose an improved multi-scale feature fusion scheme based on the classical FPN and Path Aggregation Network (PANet) architectures, aiming to enhance the model's detection capability for multi-scale objects, particularly small targets. Specifically, we have optimized the existing structure in the following aspects:

**Bidirectional Feature Propagation Mechanism :** Building upon the traditional FPN, we introduce bidirectional feature propagation paths, including both top-down and bottom-up pathways. The top-down path propagates high-level semantic features from deeper layers to shallower layers through upsampling operations, while the bottom-up path transfers high-resolution features from shallower layers to deeper layers via downsampling. This bidirectional design enables a more comprehensive fusion of features at different scales, enhancing the model's ability to represent multi-scale objects.



**FIGURE 4.** (a) FPN introduces a top-down pathway to fuse multi-scale features from level 3 to 7 (P3 - P7); (b) PANet adds an additional bottom-up pathway on top of FPN; (c) is BiFPN with better accuracy and efficiency trade-offs.

**TABLE 2.** Original YOLOv8 network structure

| Layer | f            | n | Params | Module   |
|-------|--------------|---|--------|----------|
| 0     | -1           | 1 | 464    | Conv     |
| 1     | -1           | 1 | 4672   | Conv     |
| 2     | -1           | 1 | 7360   | C2f      |
| 3     | -1           | 1 | 18560  | Conv     |
| 4     | -1           | 2 | 49664  | C2f      |
| 5     | -1           | 1 | 73984  | Conv     |
| 6     | -1           | 2 | 197632 | C2f      |
| 7     | -1           | 1 | 295424 | Conv     |
| 8     | -1           | 1 | 460288 | C2f      |
| 9     | -1           | 1 | 164608 | SPPF     |
| 10    | -1           | 1 | 0      | Upsample |
| 11    | [-1, 6]      | 1 | 0      | Concat   |
| 12    | -1           | 1 | 148224 | C2f      |
| 13    | -1           | 1 | 0      | Upsample |
| 14    | [-1, 4]      | 1 | 0      | Concat   |
| 15    | -1           | 1 | 37248  | C2f      |
| 16    | -1           | 1 | 36992  | Conv     |
| 17    | [-1, 12]     | 1 | 0      | Concat   |
| 18    | -1           | 1 | 123648 | C2f      |
| 19    | -1           | 1 | 147712 | Conv     |
| 20    | [-1, 9]      | 1 | 0      | Concat   |
| 21    | -1           | 1 | 493056 | C2f      |
| 22    | [15, 18, 21] | 1 | 752677 | Detect   |

**Multi-Level Feature Fusion and Enhancement :** We employ multi-level upsampling and downsampling operations, combined with concatenation and custom modules (e.g., C2f), to achieve effective feature fusion. Compared to the traditional FPN, this design not only enriches the detailed information of low-resolution features but also further improves the model's detection capability by concatenating high-resolution feature maps. In particular, an additional upsampling operation is introduced between P3 and P4, upsampling P3 to a higher-resolution P2 and concatenating it with the corresponding high-resolution feature map.

**Cyclic Feature Processing Strategy :** Unlike the unidirectional feature flow in FPN or BiFPN, we adopt a cyclic upsampling and downsampling strategy for feature map pro-

cessing. Specifically, high-level feature maps are progressively downsampled back to low-level feature maps, and multiple concatenation and fusion operations are performed to further enhance the interaction of features at different scales. This design strengthens the network's expressive power, particularly excelling in handling multi-scale objects.

**Lightweight and Efficiency :** We reduce computational overhead by limiting the number of bidirectional iterations and employing lightweight feature fusion modules (e.g., C2f). This ensures high performance while maintaining computational efficiency, making the model more suitable for resource-constrained practical applications.

In summary, the proposed method significantly improves the model's performance in multi-scale object detection through bidirectional feature propagation, multi-level feature fusion, cyclic feature processing, and enhanced small target detection capability.

After optimizing YOLOv8, the depth and scale of the network were increased. The original YOLOv8 had a computational load of 8.1 Giga Floating-point Operations Per second (GFLOPs), while the optimized YOLOv8 reached 11.7 GFLOPs. Although this represents an increase in computational load, it remains within an acceptable range. Despite the additional computational overhead, the enhanced network depth allows the model to learn more complex and abstract features of electric bikes, thereby improving the algorithm's accuracy. Table 2 and Table 4 present the network parameters of YOLOv8 before and after optimization. In these tables, 'f' denotes the input layer, '-1' indicates input from the previous layer, 'n' represents the connection between the current layer and the input layer, and 'Params' signifies the number of parameters in the network.

**Computational Cost Analysis:** In order to further analyze the computational cost introduced by the proposed cyclic feature processing strategy, we compared the network structure and complexity before and after modification. As shown in

**TABLE 3. Computational and Performance Comparison of Feature Fusion Variants under YOLOv8n**

| Variant         | GFLOPs | Params (M) | FPS  | Helmet AP (%) |
|-----------------|--------|------------|------|---------------|
| F-YOLOv8 (Ours) | 11.7   | 2.88       | 2000 | <b>84.6</b>   |
| BiFPN-YOLOv8    | 12.4   | 3.12       | 1760 | 82.3          |
| PANet-YOLOv8    | 12.0   | 3.01       | 1830 | 81.7          |

Table 2 and Table 4, the original YOLOv8n structure has 8.1 GFLOPs, while the improved version (F-YOLOv8) reaches 11.7 GFLOPs. Despite a 44.4% increase in GFLOPs, the model achieves significant performance improvements on small object detection tasks such as *helmet* (AP: 84.6%) and *manned* detection (AP: 67.9%), as shown in Table 7.

**Comparative Analysis with BiFPN and PANet:** To verify the cost-effectiveness of the proposed cyclic feature processing strategy, we implemented two baseline variants by replacing the cyclic fusion structure with BiFPN and PANet modules, respectively, under the same YOLOv8n backbone. All models were trained under the same conditions. The comparison results are shown in Table 3.

As shown, our cyclic fusion design achieves better AP on small-object tasks while maintaining a relatively lower parameter count and higher FPS. Although the GFLOPs slightly increase compared to the original YOLOv8n, the accuracy improvements justify the computational trade-off.

**TABLE 4. Optimized YOLOv8 network structure**

| Layer | f                | n | Params | Module   |
|-------|------------------|---|--------|----------|
| 0     | -1               | 1 | 464    | Conv     |
| 1     | -1               | 1 | 4672   | Conv     |
| 2     | -1               | 1 | 7360   | C2f      |
| 3     | -1               | 1 | 18560  | Conv     |
| 4     | -1               | 2 | 49664  | C2f      |
| 5     | -1               | 1 | 73984  | Conv     |
| 6     | -1               | 2 | 197632 | C2f      |
| 7     | -1               | 1 | 295424 | Conv     |
| 8     | -1               | 1 | 460288 | C2f      |
| 9     | -1               | 1 | 164608 | SPPF     |
| 10    | -1               | 1 | 0      | Upsample |
| 11    | [-1, 6]          | 1 | 0      | Concat   |
| 12    | -1               | 1 | 148224 | C2f      |
| 13    | -1               | 1 | 0      | Upsample |
| 14    | [-1, 4]          | 1 | 0      | Concat   |
| 15    | -1               | 1 | 37248  | C2f      |
| 16    | -1               | 1 | 0      | Upsample |
| 17    | [-1, 2]          | 1 | 0      | Concat   |
| 18    | -1               | 1 | 9408   | C2f      |
| 19    | 15               | 1 | 36992  | Conv     |
| 20    | [-1, 12]         | 1 | 0      | Concat   |
| 21    | -1               | 1 | 123648 | C2f      |
| 22    | -1               | 1 | 147712 | Conv     |
| 23    | [-1, 9]          | 1 | 0      | Concat   |
| 24    | -1               | 1 | 493056 | C2f      |
| 25    | [18, 15, 21, 24] | 1 | 618156 | Detect   |

## VII. EXPERIMENTS AND RESULT ANALYSIS

### A. EXPERIMENT PREPARATION AND SETUP

#### 1) Dataset Collection

We constructed a high-quality dataset for the safety detection of electric bike riding. The data was primarily collected in Guangzhou and Foshan, Guangdong Province. These cities



**FIGURE 5. Partial samples of dataset.**

were selected to enhance the diversity and representativeness of the dataset by covering different urban traffic environments, riding habits, and geographical characteristics. To further improve model robustness, the dataset also includes scenes under varying lighting and weather conditions. High-quality cameras were used throughout the recording process to capture clear, high-resolution image data, thereby providing a reliable foundation for subsequent model training, as shown in Fig. 5.

To ensure the quality of annotations, we employed the LabelImg image annotation tool and also organized two professionally trained annotators. Detailed annotation guidelines were established, and multiple rounds of cross-validation were conducted to ensure the consistency and accuracy of the annotations. Each image was independently annotated by two annotators, and any discrepancies were resolved by a third annotator to finalize the annotation results.

We define the following label categories:

- 1) Ebike (Rider): Used to mark the presence of a rider in the image, with the bounding box encompassing the entire rider, even if partially occluded.
- 2) Plate (License Plate): Used to mark the license plate area of the electric bike.
- 3) place (Plate Region): Used to mark the region associated with the license plate, such as "Guangzhou", "Foshan". If the license plate information is unclear or unreadable, it is labeled as "nplate."
- 4) Helmet (Wearing Helmet): If the rider is wearing a helmet, the helmet is labeled in the head region.

- 5) Nohelmet (Not Wearing Helmet): If the rider is not wearing a helmet, the head region is labeled as "nohelmet." This label is mutually exclusive with the "helmet" label, ensuring that each rider's head is labeled as either "helmet" or "nohelmet."
- 6) Manned (Carrying Passenger): If the electric bike is carrying a passenger on the rear seat, the passenger area is labeled as "manned." "Carrying a passenger" is defined as having a passenger on the rear seat, regardless of age.

Through the above data collection and annotation process, we have established a high-quality dataset suitable for electric bike riding safety detection, encompassing a wide range of riding scenarios and violations. This dataset effectively supports model training and evaluation, providing a solid data foundation for subsequent model development and performance assessment.

The distribution of the center points and sizes of the target bounding boxes in the dataset is visualized in Fig.6. Fig.6a shows the distribution of the center point coordinates with the size of the bounding boxes normalized. Data points are distributed throughout the coordinate area, indicating that the targets can appear at various locations within the image. However, the center region is noticeably darker, with coordinates approximately in the range of  $x$  between 0.3 and 0.6, and  $y$  between 0.3 and 0.7, where a large number of center points are concentrated. The image's edge regions are lighter in color, suggesting a lower probability of target center points appearing in these areas. In Fig.6b, the width and height represent the ratio of the bounding box's width and height to the width and height of the image. Data points are mainly concentrated in the lower-left corner, where the width and height ratios are smaller, indicating that most of the bounding boxes in the dataset are relatively small compared to the image size, meaning the dataset predominantly contains small targets. As the width and height ratios increase, the data points become more sparse but still present, suggesting the existence of some larger targets. Additionally, the distribution of data points shows a certain diagonal trend, indicating a positive correlation between the width and height of the bounding boxes; that is, bounding boxes with a larger width generally also have a relatively larger height.

## 2) Experimental Environment

The experimental environment is based on Windows 11, with the deep learning framework utilizing the PyTorch 2.0.0 architecture. The NVIDIA GeForce RTX 4090 D is used as the graphics card, with a video memory size of 24GB and a system memory size of 64GB. The specific experimental configuration is shown in Table 5.

## B. EXPERIMENTAL RESULTS

### 1) Training Results

To evaluate the effectiveness of the proposed F-YOLOv8 model, we conducted a comparative training experiment un-

**TABLE 5. Experimental environment configuration.**

| Parameter          | Configuration                           |
|--------------------|-----------------------------------------|
| CPU                | 15 vCPU Intel(R) Xeon(R) Platinum 8474C |
| GPU                | NVIDIA GeForce RTX 4090 D               |
| Python Version     | Python3.8.10                            |
| System environment | Windows 11                              |
| CUDA Version       | CUDA11.8                                |

der identical system environments and hyperparameter settings. YOLOv8, YOLOv9, YOLOv11, and F-YOLOv8 were all trained on the same dataset used in this study. During training, evaluation metrics including mAP@0.5, precision, and recall were recorded and averaged every 100 iterations across four epoch intervals: 200–299, 300–399, 400–499, and 500–600. The comparative results are summarized in Table 6.

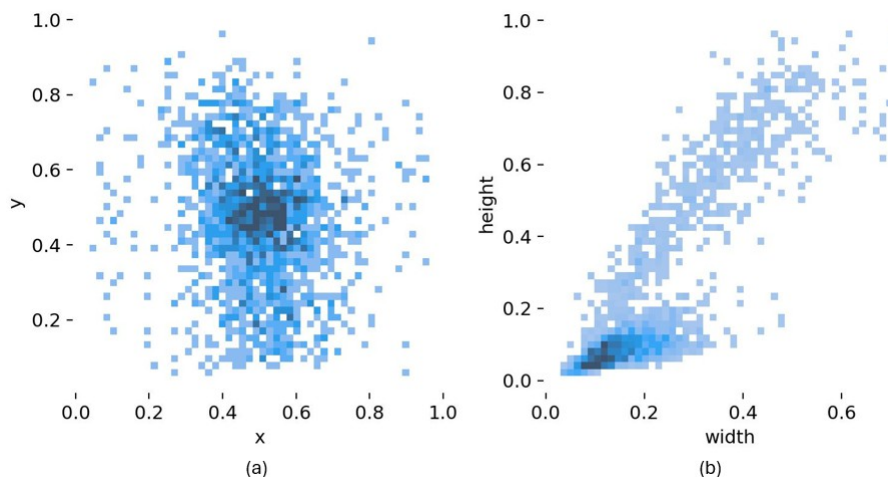
As shown, F-YOLOv8 consistently achieves competitive or superior performance across all metrics. Specifically, it maintains a high mAP@0.5 score, comparable to YOLOv9 and outperforming YOLOv8 and YOLOv11 throughout the training process. In terms of precision, F-YOLOv8 demonstrates a clear upward trend, surpassing all baseline models and reaching 88.7% in the final interval. Although its recall starts relatively low, it improves rapidly over time, closing the gap with YOLOv9 by the final epoch range. These results highlight the effectiveness and stability of the cyclic feature processing design adopted in F-YOLOv8.

To further verify the effectiveness of the F-YOLOv8 model in detecting helmets, license plates, and passenger carrying status, the trained model was subjected to field testing. The test results, shown in Fig.7, indicate that F-YOLOv8 achieves outstanding multi-object detection performance across various traffic environments. Notably, it can accurately detect small, distant targets and those that are partially occluded, thereby meeting the complex detection requirements of real-world traffic scenarios.

To further evaluate the effectiveness of the F-YOLOv8 model in static license plate detection, a series of targeted experiments were conducted. The model was tested under various conditions, including different lighting environments, angles, and distances, to ensure a comprehensive assessment of its capabilities. Static license plates, often used in parking lots, toll booths, and security checkpoints, present unique challenges such as reflections, obstructions, and font variations. The F-YOLOv8 model demonstrated robust performance, accurately identifying and localizing static license plates even in suboptimal conditions. As shown in Fig.8. This capability is crucial for applications requiring high precision and reliability, such as automated parking systems, traffic monitoring, and law enforcement. The results of these tests underscore the model's adaptability and effectiveness in handling the specific demands of static license plate detection tasks.

Fig.9 to Fig.12 demonstrate the outstanding performance





**FIGURE 6.** Dataset analysis results. (a) Location distribution of dataset target center point. (b) Distribution of dataset target size.

**TABLE 6.** Model comparison during training process.

| Epochs    |          | 200~299 | 300~399 | 400~499 | 500~600 |
|-----------|----------|---------|---------|---------|---------|
| mAP0.5    | YOLOv8   | 81.4%   | 83.9%   | 85.7%   | 86.3%   |
|           | YOLOv9   | 83.1%   | 84.5%   | 85.4%   | 86.6%   |
|           | YOLOv11  | 81.8%   | 82.5%   | 82.9%   | 83.2%   |
|           | F-YOLOv8 | 83.4%   | 84.9%   | 85.2%   | 85.9%   |
| precision | YOLOv8   | 85.4%   | 85.9%   | 86.6%   | 87.4%   |
|           | YOLOv9   | 82.7%   | 83.5%   | 84.1%   | 84.4%   |
|           | YOLOv11  | 83.7%   | 84.2%   | 84.6%   | 85.0%   |
|           | F-YOLOv8 | 85.3%   | 86.4%   | 87.4%   | 88.7%   |
| recall    | YOLOv8   | 82.3%   | 84.6%   | 84.9%   | 85.0%   |
|           | YOLOv9   | 85.5%   | 85.9%   | 86.7%   | 87.9%   |
|           | YOLOv11  | 82.4%   | 82.7%   | 83.1%   | 83.7%   |
|           | F-YOLOv8 | 79.9%   | 82.1%   | 84.7%   | 86.9%   |

and high reliability of the proposed F-YOLOv8 model across both Foshan and Guangzhou datasets through the F1-Confidence and Precision-Confidence curves. As shown in Fig.10 and Fig.11, the F1 scores remain consistently high across a wide range of confidence thresholds, indicating that the model achieves a well-balanced trade-off between precision and recall. Notably, the peak F1 values reach 0.95 and 0.80 in Foshan and Guangzhou, respectively, reflecting the robustness of F-YOLOv8 under varied urban traffic conditions. Similarly, Fig.9 and Fig.12 reveal that the model maintains high precision levels as confidence increases, with precision reaching up to 1.00 in both datasets. This indicates that high-confidence predictions made by the model are highly accurate and trustworthy. The consistently smooth and upward trends in both sets of curves further confirm the effectiveness of the confidence estimation mechanism in F-YOLOv8, supporting its applicability in safety-critical intelligent transportation scenarios.

## 2) Comparative Experiment

To comprehensively evaluate the performance of the proposed F-YOLOv8 model in electric bike safety detection, we conducted comparative experiments using a custom-built dataset, including key object categories such as electric bikes (Ebike), license plates (Plate), helmets (Helmet), and passengers (Manned). Under identical experimental settings, F-YOLOv8 was compared with multiple classic single-stage detection models (YOLOv3–YOLOv11) and the Transformer-based DETR framework. Notably, DETR was included in the comparison due to its use of global attention mechanisms, which enable modeling of long-range dependencies without relying on region proposals or post-processing steps.

Table 7 and Table 8 present the results on the Guangzhou and Foshan datasets, respectively. Across both datasets, F-YOLOv8 achieves strong and balanced performance in terms of AP, mAP@0.5, and recall.

In the Guangzhou dataset, F-YOLOv8 delivers high AP values for Ebike (96.1%) and Plate (92.9%), comparable to





FIGURE 7. Detection results of F-YOLOv8.

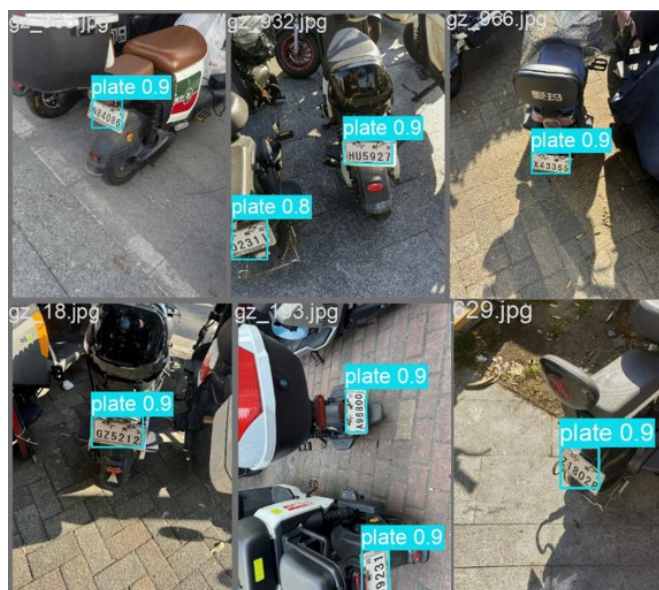


FIGURE 8. Detection results of F-YOLOv8.

or even exceeding other YOLO variants. For Helmet and Manned detection, it achieves 84.6% and 67.9%, respectively, demonstrating stable performance across varied tasks. While some YOLO variants slightly surpass F-YOLOv8 in specific categories, F-YOLOv8 achieves a favorable trade-off between detection accuracy (mAP@0.5: 83.8%), recall (0.783), and computational efficiency (2.88M parameters, 11.7 GFLOPs, and real-time FPS up to 2000).

In contrast, DETR shows significantly lower detection accuracy across all classes, with AP scores of 73.7% (Ebike), 47.1% (Plate), 56.3% (Helmet), and 39.2% (Manned). Although its GFLOPs is relatively low (6.67), its extremely limited FPS (10.58) makes it unsuitable for real-time safety

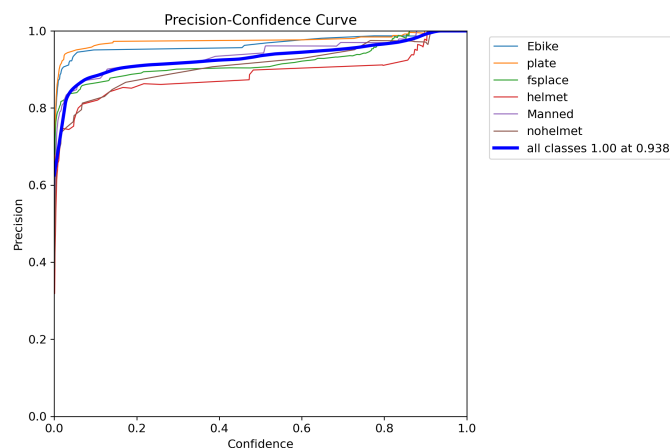


FIGURE 9. Precision-Confidence Curve of F-YOLOv8 on Foshan Dataset.

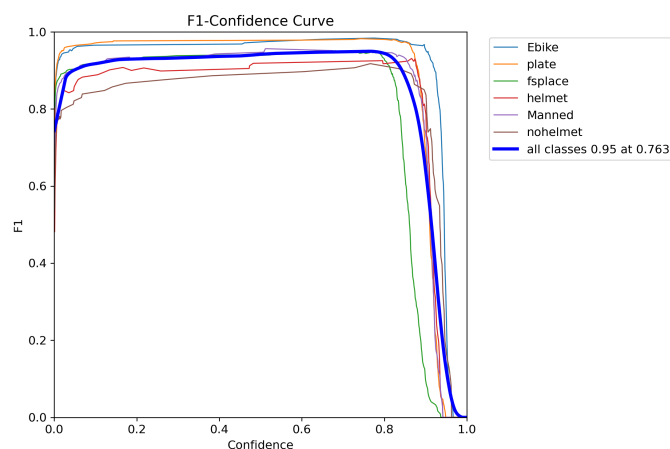


FIGURE 10. F1-Confidence Curve of F-YOLOv8 on Foshan Dataset.

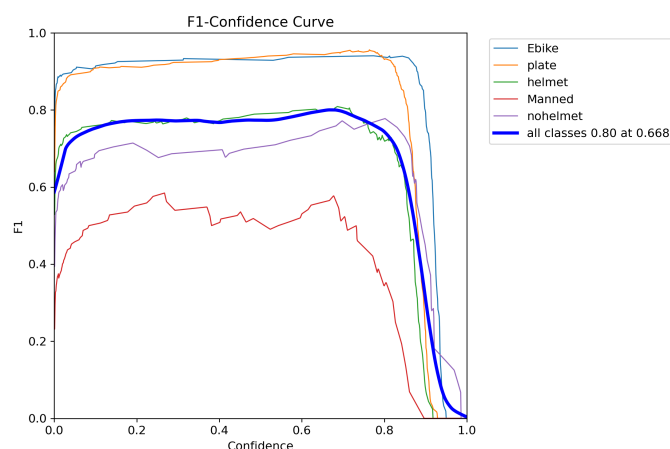
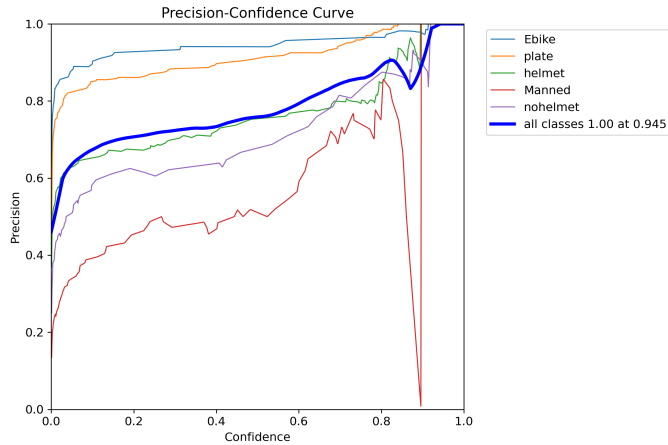


FIGURE 11. F1-Confidence Curve of F-YOLOv8 on Guangzhou Dataset

detection.

In the Foshan dataset, which introduces more varied light-



**FIGURE 12.** Precision-Confidence Curve of F-YOLOv8 on Guangzhou Dataset

ing and weather conditions, F-YOLOv8 again demonstrates robust generalization. It achieves 98.6% and 98.4% AP for Ebike and Plate, along with 90.9% and 97.0% AP for Helmet and Manned detection, respectively. Its overall mAP@0.5 reaches 96.6%, matching or exceeding that of competing YOLO models while maintaining a lightweight model size and low computational cost.

These results validate the effectiveness of our architectural enhancements and also justify the choice of YOLOv8 as the foundation of our model. Although the YOLO series has advanced to version 11, we selected YOLOv8 due to its stability, maturity, and suitability for the specific demands of our task. YOLOv8 adopts an anchor-free detection head and a streamlined architecture, which are particularly advantageous for identifying small and densely packed objects such as helmets and license plates. While newer versions like YOLOv9–YOLOv11 introduce additional complexity, they have not yet demonstrated consistent advantages in lightweight, real-time scenarios. Furthermore, YOLOv8's modular structure facilitates the integration of our cyclic feature processing mechanism, and its well-established open-source ecosystem ensures high reproducibility and deployment efficiency.

In conclusion, F-YOLOv8 consistently outperforms DETR and achieves competitive results compared to other YOLO variants, while maintaining an excellent trade-off between detection performance and inference speed. These findings demonstrate the robustness and practicality of F-YOLOv8 in real-time electric bike safety applications under diverse environmental conditions.

### 3) Ablation Study

To verify the effectiveness of the proposed components in F-YOLOv8, we conduct a series of ablation experiments as summarized in Table 9. The original YOLOv8n model is used as the baseline (A0), and the full version of our method with both the P2 high-resolution branch and cyclic feedback

structure is denoted as A1. We then selectively remove key components to create two variants: A2 retains only the high-resolution P2 branch, while A3 preserves only the PAN-like feedback structure.

Compared to the baseline A0, the full F-YOLOv8 model (A1) achieves substantial performance gains in both detection accuracy and robustness. Specifically, A1 improves the mAP@0.5 from 0.843 to 0.966 and the mAP@0.5:0.95 from 0.404 to 0.812, demonstrating the combined effectiveness of the proposed architectural enhancements.

To disentangle the contributions of each component, we analyze A2 and A3 in detail:

- 1) A2 (Only P2 Branch): This variant introduces the high-resolution P2 feature without the cyclic feedback mechanism. Although the model's complexity is significantly reduced (1.81M parameters vs. 3.10M), it still achieves a promising mAP@0.5 of 0.921. This suggests that incorporating high-resolution shallow features greatly benefits small object detection, especially for targets like helmet and ebike. However, the mAP@0.5:0.95 drops to 0.614, indicating that the absence of deeper semantic reinforcement (as provided by the feedback path) limits its overall precision across IoU thresholds.
- 2) A3 (Only PAN Feedback): By contrast, this variant retains the feedback structure but removes the P2 branch. The performance drops dramatically (mAP@0.5: 0.455; mAP@0.5:0.95: 0.123), confirming that without high-resolution features, the model struggles with fine-grained detection tasks. This result emphasizes that simply propagating features back through PAN is insufficient if the original spatial detail is lost in the early stages.

In summary, the full F-YOLOv8 (A1) benefits from the complementary roles of both the P2 branch and the feedback path: the former enriches spatial resolution for small objects, while the latter enhances semantic flow and consistency across feature levels. The combination yields significant improvements over both the baseline and the ablated variants.

## VIII. CONCLUSIONS

This paper proposes a MTL framework that integrates license plate region recognition, enabling simultaneous license plate recognition, identification of the license plate's region, helmet wearing detection, and carrying passenger detection. By incorporating the Transformer architecture, the model enhances the feature representation capability of YOLO, improving its adaptability to complex scenarios and performance in detecting small objects. Additionally, the introduction of the FPN further enhances multi-scale object detection capabilities. Compared to YOLOv8, the proposed model shows significant improvements across multiple metrics. Specifically, in object detection accuracy, the model achieves 96.1%, 92.9%, 84.6%, and 67.9% for the Ebike, Plate, Helmet, and Manned tasks, respectively, showing clear improvements over YOLOv8. In terms of mAP0.5, the proposed model achieves

**TABLE 7. GuangZhou Comparative experiment.**

| Model   | AP    |       |        |        | mAP0.5 | recall | Parameters/10 <sup>6</sup> | GFLOPs |
|---------|-------|-------|--------|--------|--------|--------|----------------------------|--------|
|         | Ebike | Plate | Helmet | Manned |        |        |                            |        |
| Detr    | 73.7% | 47.1% | 56.3%  | 39.2%  | 53.3%  | 0.337  | 5.9                        | 6.67   |
| YOLOv3  | 96.5% | 95.8% | 89.0%  | 76.9%  | 83.7%  | 0.777  | 103.67                     | 282.2  |
| YOLOv8  | 93.4% | 91.7% | 78.2%  | 66.5%  | 84.3%  | 0.862  | 3.01                       | 8.1    |
| YOLOv9  | 91.3% | 95.6% | 84.6%  | 72.6%  | 88.4%  | 0.888  | 2.61                       | 10.7   |
| YOLOv10 | 86.8% | 85.7% | 81.0%  | 76.5%  | 81.1%  | 0.734  | 2.69                       | 8.2    |
| YOLOv11 | 95.0% | 94.0% | 77.1%  | 61.2%  | 86.1%  | 0.851  | 2.58                       | 6.3    |
| Ours    | 96.1% | 92.9% | 84.6%  | 67.9%  | 83.8%  | 0.783  | 2.88                       | 11.7   |

**TABLE 8. Foshan Comparative experiment.**

| Model   | AP    |       |        |        | mAP0.5 | recall | Parameters/10 <sup>6</sup> | GFLOPs |
|---------|-------|-------|--------|--------|--------|--------|----------------------------|--------|
|         | Ebike | Plate | Helmet | Manned |        |        |                            |        |
| Detr    | 58.1% | 45.8% | 39.6%  | 23.5%  | 44.2%  | 0.316  | 5.9                        | 6.67   |
| YOLOv3  | 97.3% | 97.9% | 90.0%  | 97.8%  | 96.2%  | 0.884  | 103.67                     | 282.2  |
| YOLOv8  | 95.5% | 98.5% | 93.9%  | 96.6%  | 96.9%  | 0.947  | 3.01                       | 8.1    |
| YOLOv9  | 98.1% | 98.2% | 91.2%  | 91.5%  | 96.6%  | 0.943  | 2.61                       | 10.7   |
| YOLOv10 | 97.7% | 98.5% | 92.5%  | 96.8%  | 95.9%  | 0.925  | 2.69                       | 8.2    |
| YOLOv11 | 96.2% | 99.0% | 89.0%  | 94.0%  | 93.0%  | 0.956  | 2.58                       | 6.3    |
| Ours    | 98.6% | 98.4% | 90.9%  | 97.0%  | 96.6%  | 0.941  | 2.88                       | 11.7   |

**TABLE 9. Ablation study of each component in the proposed F-YOLOv8.**

| ID | Configuration               | Params (M) | GFLOPs | mAP@0.5 | mAP@0.5:0.95 | FPS   |
|----|-----------------------------|------------|--------|---------|--------------|-------|
| A0 | Original YOLOv8n (Baseline) | 3.20       | 8.7    | 0.843   | 0.404        | 3333  |
| A1 | Full F-YOLOv8 (Ours)        | 3.10       | 11.7   | 0.966   | 0.812        | 192   |
| A2 | Only P2 branch added        | 1.81       | 10.5   | 0.921   | 0.614        | 2.49  |
| A3 | Only PAN feedback path      | 3.01       | 8.1    | 0.455   | 0.123        | 10.16 |

83.8%, slightly higher than YOLOv8's 84.3%. Although the improvement in mAP0.5 is not significant, enhancements in other key metrics indicate superior overall performance within the MTL framework. In terms of model efficiency, F-YOLOv8 reduces the parameter count by 4.3%, effectively lowering the model's storage requirements. Although GFLOPs increase by 44.4%, this results in a substantial improvement in FPS, reaching 80% of YOLOv8's performance, with faster detection speeds. Moreover, compared to other multi-task detection models, F-YOLOv8 excels in small object detection and adaptation to complex scenarios, making it better suited for resource-constrained edge computing devices, with broad application prospects in intelligent transportation, security monitoring, and other fields.

However, due to the influence of many complex factors in real-world scenarios, there are still some limitations in the detection algorithm proposed in this paper, which warrant further investigation:

- 1) In practical applications, the background is often complex and dynamic, which may lead to false positives or missed detections by the model. Specifically, in tasks

like license plate recognition and helmet detection, complex backgrounds may interfere with the model's judgment, affecting detection accuracy.

- 2) Although Transformer and Feature Pyramid Networks are introduced to improve small object detection capability, there is still room for improvement in the model's detection performance in extreme cases (e.g., when the object is very small or the resolution is extremely low).
- 3) For future work, we will delve into and optimize the detection of electric bikes in different regions, considering variations in usage habits, license plate formats, and traffic regulations across regions. We will optimize the model based on these regional characteristics to ensure its detection accuracy and adaptability in various regions.

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